DNS Server Configuration

Q1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

Solution: DNS Server Setup

BIND is DNS server software used to manage DNS zones and records.

Steps:

1. On a Linux machine with administrator access, install BIND:

sudo apt update

sudo apt install bind9 bind9-utils dnsutils

2. Open the configuration file:

sudo nano /etc/bind/named.conf.local

3. Add the following zone configuration:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

4. Create the zone file:

sudo nano /etc/bind/db.musicmania.in

Result: The musicmania.in DNS zone is defined in the BIND configuration.

Q2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Solution : Add an A record for www.musicmania.in

Add the following records to /etc/bind/db.musicmania.in:

```
$TTL 86400
@ IN SOA ns1.musicmania.in. admin.musicmania.in. (
    2026102001
    3600
    1800
    604800
    86400
)
@    IN NS ns1.musicmania.in.
ns1  IN A  192.168.1.50
www  IN A  192.168.1.50
```

The A record maps www.musicmania.in to 192.168.1.50.

Verify using nslookup:

```
nslookup www.musicmania.in 127.0.0.1
```

Or use:

```
dig @127.0.0.1 www.musicmania.in A
```

Expected result: The DNS answer contains 192.168.1.50, provided the server is configured and running correctly.

Q3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Solution : Add a CNAME record for play.musicmania.in

Add this line to the zone file:

play IN CNAME www.musicmania.in.

Restart or reload BIND after validating the configuration.

Test the record:

nslookup play.musicmania.in 127.0.0.1

Expected result: play.musicmania.in resolves through the alias www.musicmania.in to 192.168.1.50.

Explanation: A CNAME record makes one domain name an alias of another domain name.

Q4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Hint : Include at least three different troubleshooting commands or checks in your documentation.

Solution : Troubleshoot DNS resolution problems

If musicmania.in does not resolve correctly, I would follow these steps.

Step 1: Check the configuration syntax

sudo named-checkconf

This checks the BIND configuration for syntax errors.

Step 2: Validate the zone file

sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in

This checks the zone file for errors.

Step 3: Restart the DNS service

sudo systemctl restart bind9

Step 4: Check the service status

sudo systemctl status bind9

This shows whether the DNS service is running.

Step 5: Test DNS resolution

dig @127.0.0.1 www.musicmania.in

Alternatively : nslookup www.musicmania.in 127.0.0.1

Step 6: Check firewall settings

Ensure DNS traffic on port 53 is permitted over UDP and TCP where required, and that the server is listening on the expected interface.

Common problems and solutions

Problem	Solution
Incorrect zone syntax	Run named-checkconf and correct errors.
Invalid zone file	Run named-checkzone and fix the reported issue.
DNS service stopped	Restart BIND and check its status.
Incorrect IP record	Correct the A record in the zone file.
DNS traffic blocked	Check firewall rules and port 53 access.